



Curriculum for Undergraduate Degree (B.Tech.) in Mechanical Engineering (w.e.f. AY: 2020-21)

Part II: Course Structure & Semester wise Credit Distribution

First Semester (First Year)							
Mandatory Induction Program- 3 Weeks Duration							
Sl No.	Category	Course Code	Course Name	Contact Hours			Credits
				L	T	P	
Theory							
1	BS	BS-CH101	Chemistry	3	1	0	4
2	BS	BS-M101	Mathematics-I	3	1	0	4
3	ES	ES-CS101	Programming for Problem Solving	3	0	0	3
Total Theory				9	2	0	11
Practical/ Sessional							
1	BS	BS-CH191	Chemistry Laboratory	0	0	3	1.5
2	ES	ES-CS191	Programming for Problem Solving Lab	0	0	4	2
3	ES	ES-ME192	Workshop/ Manufacturing Practices	1	0	4	3
4	HM	HM-HU191	Language Laboratory	0	0	2	1
Total Practical/ Sessional				1	0	13	7.5
Total of First Semester				10	2	13	18.5

Second Semester (First Year)							
Sl No.	Category	Course Code	Course Name	Contact Hours			Credits
				L	T	P	
Theory							
1	BS	BS-PH201	Physics	3	1	0	4
2	BS	BS-M201	Mathematics-II	3	1	0	4
3	ES	ES-EE201	Basic Electrical and Electronics Engineering	3	1	0	4
4	HM	HM-HU201	English	2	0	0	2
	Total Theory			11	3	0	14
Practical/ Sessional							



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1	BS	BS-PH291	Physics Laboratory	0	0	3	1.5
2	ES	ES-EE291	Basic Electrical & Electronics Engineering Lab	0	0	4	2
3	ES	ES-ME291	Engineering Graphics & Design	1	0	4	3
<i>Total Practical/ Sessional</i>				1	0	10	6.5
Total of Second Semester				13	3	10	20.5

Third Semester (Second Year)							
Sl No.	Category	Course Code	Course Name	Contact Hours			Credits
				L	T	P	
Theory							
1	BS	BS-M 303	Mathematics-III	2	1	0	3
2	BS	BS-BIO 301	Biology	2	0	0	2
3	ES	ES-ME 301	Materials Engineering	3	0	0	3
4	ES	ES-ME 302	Engineering Mechanics	3	1	0	4
5	PC	PC-ME 301	Thermodynamics	3	1	0	4
6	PC	PC-ME 302	Basic Manufacturing Processes	4	0	0	4
	Total Theory			17	3	0	20
Practical/ Sessional							
1	PC	PC-ME 391	Basic Manufacturing Processes Lab	0	0	3	1.5
	Total Practical/ Sessional			0	0	3	1.5
	Total of Third Semester			17	3	3	21.5

Fourth Semester (Second Year)							
Sl No.	Category	Course Code	Course Name	Contact Hours			Credits
				L	T	P	
Theory							
1	BS	BS- M 404	Numerical methods	2	0	0	2
2	PC	PC-ME 401	Applied Thermodynamics	3	1	0	4
3	PC	PC-ME 402	Fluid Mechanics and Fluid Machines	3	1	0	4



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4	PC	PC-ME 403	Strength of Materials	3	1	0	4
5	PC	PC-ME 404	Metrology and Instrumentation	4	0	0	4
<i>Total Theory</i>				15	3	0	18
Practical/ Sessional							
1	PC	PC-ME 491	Metrology and Instrumentation Lab	0	0	3	1.5
2	PC	PC-ME 492	Machine Drawing-I	0	0	3	1.5
3	MC	MC471	Environmental Sciences	2	0	0	0
<i>Total Practical/ Sessional</i>				0	0	6	3
Total of Fourth Semester				17	3	6	21

Fifth Semester (Third Year)							
Sl No.	Category	Course Code	Course Name	Contact Hours			Credits
				L	T	P	
Theory							
1	HM	HM-HU 503	Effective Technical Communication	3	0	0	3
2	PC	PC-ME 501	Heat Transfer	3	1	0	4
3	PC	PC-ME 502	Solid Mechanics	3	1	0	4
4	PC	PC-ME 503	Kinematics and Theory of Machines	3	1	0	4
	Total Theory			12	3	0	15
Practical/ Sessional							
1	PC	PC-ME591	Mechanical Engineering Laboratory-I (Thermal)	0	0	3	1.5
2	PC	PC-ME 592	Machine Drawing-II	0	0	3	1.5
3	PW	PW-ME 581	Project-I (30 hrs. Total)	0	0	2	1
5	MC	MC 571	Aptitude Skill Development-I	2	0	0	0
	Total Practical/ Sessional			0	0	8	4
	Total of Fifth Semester			14	3	8	19

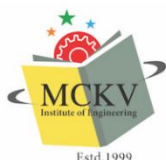


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Sixth Semester (Third Year)							
Sl No.	Category	Course Code	Course Name	Contact Hours			Credits
				L	T	P	
Theory							
1	PC	PC-ME 601	Manufacturing Technology	4	0	0	4
2	PC	PC-ME 602	Design of Machine Elements	3	1	0	4
3	PE	PE-ME 601	Elective-I	3	0	0	3
4	PE	PE-ME 602	Elective-II	3	0	0	3
5	HM	HM-HU 603	Operations Research	4	0	0	4
	Total Theory			17	1	0	18
Practical/ Sessional							
1	PC	PC-ME 691	Mechanical Engineering Laboratory-II (Design)	0	0	3	1.5
2	PW	PW-ME 681	Project-II (60 hrs. Total)	0	0	4	2
3	HM	HM-HU 691	Soft Skill Development Lab	0	0	2	1
4	MC	MC 671	Aptitude Skill Development-II	2	0	0	0
5	MC	MC 672	Constitution of India	2	0	0	0
	Total Practical/ Sessional			0	0	9	4.5
	Total of Sixth Semester			21	1	9	22.5

Seventh Semester (Fourth Year)							
Sl No.	Category	Course Code	Course Name	Contact Hours			Credits
				L	T	P	
Theory							
1	PC	PC-ME 701	Advanced Manufacturing Technology	3	0	0	3
2	PE	PE-ME 701	Elective-III	3	0	0	3
3	PE	PE-ME 702	Elective-IV	3	0	0	3
4	OE	OE-AUE 701	Open Elective-I	3	0	0	3
5	HM	HM-HU 701	Economics for Engineers	3	0	0	3
	Total Theory			15	0	0	15
Practical/ Sessional							



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1	PC	PC-ME 791	Mechanical Engineering Laboratory III (Manufacturing)	0	0	3	1.5
2	PW	PW-ME 781	Project-III	0	0	6	3
<i>Total Practical/ Sessional</i>				0	0	9	4.5
Total of Seventh Semester				15	0	9	19.5

Eighth Semester (Fourth Year)							
Sl No.	Category	Course Code	Course Name	Contact Hours			Credits
				L	T	P	
Theory							
1	PE	PE-ME 801	Elective V	3	0	0	3
2	PE	PE-ME 802	Elective VI	3	0	0	3
3	OE	OE-AUE 801	Open Elective-II	3	0	0	3
4	OE	OE-AUE 801	Open Elective-III	3	0	0	3
	Total Theory			12	0	0	12
Practical/ Sessional							
1	PW	PW-ME 881	Project-IV	0	0	8	4
2	PW	PW-ME 882	Comprehensive Viva Voce	0	0	0	1.5
	Total Practical/ Sessional			0	0	8	5.5
	Total of Seventh Semester			12	0	8	17.5

List of Professional Electives in “**Semester VI**” for (Elective-I) PE-ME601 and (Elective-II) PE-ME602

Subject Code	Subject Name
PE-ME 601A	Internal Combustion Engines and Gas Turbines
PE-ME 601B	Composite Materials
PE-ME 601C	Turbo Machinery
PE-ME 601D	Robotics
PE-ME 602A	Refrigeration and Air Conditioning
PE-ME 602B	Fluid Power Control
PE-ME 602C	Mechatronics
PE-ME 602D	Material Handling



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List of Professional Electives in “**Semester VII**” for (Elective-III) PE-ME 701 and (Elective-IV) PE-ME 702

Subject Code	Subject Name
PE-ME 701A	Automobile Engineering
PE-ME 701B	Computational Fluid Dynamics
PE-ME 701C	Non-Conventional Energy Sources
PE-ME 702A	Finite Element Analysis
PE-ME 702B	CAD/CAM
PE-ME 702C	Advanced Welding Technology
PE-ME 702D	Additive Manufacturing

List of Open Electives in “**Semester VII**” (OE-ME 701)

Subject Code	Subject Name
OE-AUE 701A	Industrial Engineering
OE-M 701B	Computational Methods in Engineering
OE-BS 701A	Biomechanics and Biomaterials
OE-IT 701J	Artificial Intelligence and Machine Learning

List of Professional Electives in “**Semester VIII**” for (Elective-V) PE-ME 801 and (Elective-VI) PE-ME 802

Subject Code	Subject Name
PE-ME 801A	Power Plant Engineering
PE-ME 801B	Energy Conservation and Management
PE-ME 801C	Cryogenics
PE-ME 802A	Tribology
PE-ME 802B	Material Testing
PE-ME 802C	Micro-machining
PE-ME 802D	Maintenance Engineering

List of Open Electives in “**Semester VIII**” (OE-ME 801 and OE-ME 802)

Subject Code	Subject Name
OE-AUE 801B	Total Quality Management
OE-HU 801K	Entrepreneurship Development
OE-AUE 801C	Analysis and Performance of Fluid Machines
OE-AUE 801D	Automation and Control